TTK4115 Assignment 1

Problem 1:

2 pt

a) $\dot{x}_1 = \dot{y}$

$= x_2 + v$

$\dot{x}_2 = \ddot{y} - \dot{v}$

$= (\dot{v} + 4v - 2\dot{y}) - \dot{v}$

$= 4v - 2\dot{y}$

$= 4v - 2(x_2 + v)$

$= \underline{2v - 2x_2}$

$$\frac{d}{dt}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 \\ 0 & -2 \end{bmatrix}}_{A}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \underbrace{\begin{bmatrix} 1 \\ 2 \end{bmatrix}}_{B} v$$

$$y = \underbrace{\begin{bmatrix} 1 & 0 \end{bmatrix}}_{C}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \underbrace{\begin{bmatrix} 0 \end{bmatrix}}_{D} v$$

b) $\hat{q} = \begin{bmatrix} 1 & 0 \end{bmatrix} \left(sI - \begin{bmatrix} 0 & 1 \\ 0 & -2 \end{bmatrix} \right)^{-1} \begin{bmatrix} 1 \\ 2 \end{bmatrix} + 0$

$= \begin{bmatrix} 1 & 0 \end{bmatrix} \left(\begin{bmatrix} s & -1 \\ 0 & s+2 \end{bmatrix}^{-1} \right) \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

$= \begin{bmatrix} 1 & 0 \end{bmatrix} \frac{1}{s(s+2)} \begin{bmatrix} s+2 & 1 \\ 0 & s \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

$= \frac{1}{s(s+2)} \begin{bmatrix} s+2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

$= \frac{s+2+2}{s(s+2)}$

$= \underline{\underline{\frac{s+4}{s(s+2)}}}$

c) $\ddot{y}(t) + 2\dot{y}(t) = \dot{u}(t) + 4u(t)$

$\downarrow$ $Y(s),\ U(s) = \mathcal{L}\{y(t)\},\ \mathcal{L}\{u(t)\}$

$y(0) = \dot{y}(0) = u(0) = 0$

$s^2 Y(s) + 2sY(s) = sU(s) + 4U(s)$

$Y(s) \cdot (s^2 + 2s) = U(s) \cdot (s+4)$

$\hat{q}(s) = \frac{Y(s)}{U(s)} = \underline{\underline{\frac{s+4}{s(s+2)}}}$

d) $\frac{A}{s} + \frac{B}{s+2} = \frac{s+4}{s(s+2)} = \hat{q}(s)$

$A(s+2) + Bs = s + 4$

$\Rightarrow \quad A + B = 1$

$\quad\quad 2A = 4$

$\Rightarrow \quad A = 2$

$\quad\quad B = -1$

$\Rightarrow \hat{q}(s) = \frac{2}{s} - \frac{1}{s+2}$

$\underline{q(t) = 2 - e^{-2t}}$

↳ Impulsantwort $v(t) = \delta(t)$

Problem 2

a) $(sI - A)^{-1}$

$$= \begin{bmatrix} s & -1 \\ 0 & s+3 \end{bmatrix}^{-1}$$

$$= \frac{1}{s(s+3)} \begin{bmatrix} s+3 & -1 \\ 0 & s \end{bmatrix}$$

$$= \begin{bmatrix} 1/s & -1/s(s+3) \\ 0 & 1/(s+3) \end{bmatrix}$$

$$e^{At} = \mathcal{L}^{-1}\{(sI - A)^{-1}\}$$

$$= \begin{bmatrix} \mathcal{L}^{-1}\{1/s\} & \mathcal{L}^{-1}\{-1/(s(s+3))\} \\ \mathcal{L}^{-1}\{0\} & \mathcal{L}^{-1}\{1/(s+3)\} \end{bmatrix}$$

$$= \begin{bmatrix} 1 & \mathcal{L}^{-1}\{\frac{1}{3}(\frac{-1}{s} + \frac{1}{s+3})\} \\ 0 & e^{-3t} \end{bmatrix}$$

$$= \begin{bmatrix} 1 & \frac{1}{3}(-1 + e^{-3t}) \\ 0 & e^{-3t} \end{bmatrix}$$

$$\frac{A}{s} + \frac{B}{s+3} = \frac{-1}{s(s+3)}$$

$$A(s+3) + Bs = -1$$

$$A + B = 0$$

$$3A = -1$$

$$A = -\frac{1}{3}$$

$$B = \frac{1}{3}$$

b) $A = \begin{bmatrix} 0 & 1 \\ 0 & -3 \end{bmatrix}$

$\Rightarrow -\lambda(-\lambda - 3) = 0$

$\Rightarrow \lambda_1 = 0, \quad \lambda_2 = -3$

$\lambda_1 = 0$

$(A - I\lambda_1)v_1 = 0$

$\Rightarrow \left[\begin{array}{cc|c} 0 & 1 & 0 \\ 0 & -3 & 0 \end{array}\right] \sim \left[\begin{array}{cc|c} 0 & 1 & 0 \\ 0 & 0 & 0 \end{array}\right] \Rightarrow k_2 = 0$

$v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

c) $\lambda_2 = -3$

$(A - I\lambda_2)v_2 = 0$

$\Rightarrow \left[\begin{array}{cc|c} +3 & 1 & 0 \\ 0 & 0 & 0 \end{array}\right] \Rightarrow -3k_1 + k_2 = 0$

$v_2 = \begin{bmatrix} 1 \\ -3 \end{bmatrix}$

$\hat{A} = \begin{bmatrix} 0 & 0 \\ 0 & -3 \end{bmatrix} \qquad Q = \begin{bmatrix} 1 & 1 \\ 0 & -3 \end{bmatrix}$

$Q^{-1} = \frac{1}{3}\begin{bmatrix} 3 & 1 \\ 0 & -1 \end{bmatrix}$

d) $e^{At} = Q e^{\hat{A}t} Q^{-1}$

$$= \begin{bmatrix} 1 & 1 \\ 0 & -3 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ 0 & e^{-3t} \end{bmatrix} \cdot \frac{1}{3} \begin{bmatrix} 3 & 1 \\ 0 & -1 \end{bmatrix}$$

$$= \frac{1}{3} \begin{bmatrix} 1 & e^{-3t} \\ 0 & -3e^{-3t} \end{bmatrix} \cdot \begin{bmatrix} 3 & 1 \\ 0 & -1 \end{bmatrix}$$

$$= \frac{1}{3} \begin{bmatrix} 3 & 1 - e^{-3t} \\ 0 & 3e^{-3t} \end{bmatrix}$$

e) $\dot{x} = Ax + Bu$

$\bar{x} = Q^{-1}x, \quad x = Q\bar{x}$

$\dot{\bar{x}} = Q^{-1}\dot{x}$

$\dot{\bar{x}} = Q^{-1}Ax + Q^{-1}Bu$

$\dot{\bar{x}} = Q^{-1}AQ\bar{x} + Q^{-1}Bu$

$\underline{\dot{\bar{x}} = \hat{A}\bar{x} + Q^{-1}Bu}$

From the general formula we get that:

$$\bar{x} = e^{\hat{A}t}x_0 + \int_0^t e^{\hat{A}(t-\tau)}Q^{-1}B\,d\tau$$

And that

$$y = Cx + Du$$

$$y = CQ\bar{x} + D$$

So:

$$y(t) = (Q e^{\hat{A}t} x_0 + CQ \int_0^t e^{\hat{A}(t-\tau)} Q^{-1} B \, d\tau + \bar{5}$$

$$y(t) = \begin{bmatrix} 3 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & -3 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & e^{-3t} \end{bmatrix} x_0 + \begin{bmatrix} 3 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & -3 \end{bmatrix} \cdot \int_0^t \begin{bmatrix} 1 & 0 \\ 0 & e^{-3t} e^{3\tau} \end{bmatrix} d\tau \cdot \frac{1}{3} \begin{bmatrix} 3 & 1 \\ 0 & -1 \end{bmatrix} \cdot \begin{bmatrix} -3 \\ 0 \end{bmatrix} + 5$$

$$y(t) = 3 \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & e^{-3t} \end{bmatrix} x_0 + 3 \int_0^t \begin{bmatrix} 1 & e^{-3t} e^{3\tau} \end{bmatrix} d\tau \cdot \frac{1}{3} \begin{bmatrix} -6 \\ 0 \end{bmatrix} + 5$$

$$y(t) = 3 \begin{bmatrix} 1 & e^{-3t} \end{bmatrix} x_0 + \begin{bmatrix} 3t & 0 \end{bmatrix} \cdot \begin{bmatrix} -2 \\ 0 \end{bmatrix} + 5$$

$$y(t) = 3 \begin{bmatrix} 1 & e^{-3t} \end{bmatrix} x_0 - 6t + 5$$

f) Using the expressions from e), we get:

$$Y(1) = \begin{bmatrix} 3 & e^{-1} \end{bmatrix} x_0 - 6 \cdot 1 + 5 = 4$$

$$\Rightarrow 3x_1 + e^{-1}x_2 = 5$$

$$Y(2) = \begin{bmatrix} 3 & e^{-2} \end{bmatrix} x_0 - 6 \cdot 2 + 5 = 4$$

$$\Rightarrow 3x_1 + e^{-2}x_2 = 11$$

Problem 3

a) $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} r \\ \dot{r} \end{bmatrix}$, $u = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$

$$\dot{x}_1 = \dot{r} = x_2$$

$$\dot{x}_2 = \frac{x_1 u_1 - 2 u_1 x_2}{\sqrt{x_1^2 + 9}} + u_2 - 3x_1 - 4$$

$$y = 5\sqrt{x_1^2 + 9}$$

$$h(x,u) = \begin{bmatrix} x_2 \\ u_1 \dfrac{x_1 - 2x_2}{\sqrt{x_1^2 + 9}} + u_2 - 3x_1 - 4 \end{bmatrix}$$

$$f(x,u) = 5\sqrt{x_1^2 + 9}$$

b) $\dot{x}(t) \sim \dot{x}_0 + \left.\frac{d\,x(t)}{dt}\right|_{x=x_0,\, u=u_0}$

$= 0 + \frac{\partial h}{\partial x}(x_0, u_0)(x - x_0) + \frac{\partial h}{\partial u}(x_0, u_0)(u - u_0)$

$= \frac{\partial h}{\partial x}(x_0, u_0)\,\bar{x} + \frac{\partial h}{\partial u}(x_0, u_0)\,\bar{u}$

$$\frac{\partial h}{\partial x}(x_0, u_0) = \begin{bmatrix} 0 & 1 \\ -\frac{1}{2}\frac{u_1(x_1 - 2x_2)}{(x_1^2+4)^{3/2}} \cdot 2x_1 + \frac{u_1}{\sqrt{x_1^2+4}} - 3 & \frac{-2u_1}{\sqrt{x_1^2+4}} \end{bmatrix}$$

$$\frac{\partial h}{\partial x}\left(\begin{bmatrix} 4 \\ 2 \end{bmatrix}, \begin{bmatrix} 5 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 0 & 1 \\ \frac{5}{\sqrt{x^2+4}} - 3 & -2 \cdot \frac{5}{\sqrt{25}} \end{bmatrix}$$

$$A = \begin{bmatrix} 0 & 1 \\ -2 & -2 \end{bmatrix}$$

$$\frac{\partial h}{\partial v}(x_0, v_0) = \begin{bmatrix} 0 & 0 \\ \frac{x_1 - 2x_2}{\sqrt{x_1^2 + q}} & 1 \end{bmatrix}$$

$$\frac{\partial h}{\partial v}\left(\begin{bmatrix} 4 \\ 2 \end{bmatrix}, \begin{bmatrix} 5 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = B$$

$$dt = \frac{\partial t}{\partial x_0}(x_0, v_0)(x - x_0) + \frac{\partial t}{\partial v}(x_0, v_0)(v - v_0)$$

$$\frac{\partial t}{\partial x_0}(x_0, v_0) = \begin{bmatrix} 5 \cdot 2 x_1 \cdot \frac{1}{2}(x_1^2 + q)^{-1/2} & 0 \end{bmatrix}$$

$$\frac{\partial t}{\partial x_0}\left(\begin{bmatrix} 4 \\ 2 \end{bmatrix}, \begin{bmatrix} 5 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} \frac{5 \cdot 4}{\sqrt{25}} & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 4 & 0 \end{bmatrix} = C$$

$$\frac{\partial f}{\partial u}(x_0, v_0) = \begin{bmatrix} 0 & 0 \end{bmatrix} = D$$

Problem 4

a) $-\lambda(-\lambda-6)-(-9)\cdot 1 = 0$

$\lambda^2 + 6\lambda + 9 = 0$

$(\lambda+3)^2 = 0$

$\lambda_{1,2} = -3$

$(A-\lambda_1)v = 0$

$$\Rightarrow \left[\begin{array}{cc|c} 3 & -9 & 0 \\ 1 & -3 & 0 \end{array}\right] \sim \left[\begin{array}{cc|c} 0 & 0 & 0 \\ 1 & -3 & 0 \end{array}\right]$$

$\Rightarrow x_1 - 3x_2 = 0$

$\Rightarrow v = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$

(4) No, because the matrix

$P = \begin{bmatrix} 3 & 3 \\ 1 & 1 \end{bmatrix}$ does not have

an inverse P^{-1}

So the similarity transform

$A = P \Lambda P^{-1}$ does not exist.

(5) The jordan block becomes:

$$J = \begin{bmatrix} -3 & 1 \\ 0 & -3 \end{bmatrix}$$

$$v_1 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

$$(A - a_2 I) v_{j1} = v_1$$

$$\begin{bmatrix} 3 & -9 & 3 \\ 1 & -3 & 1 \end{bmatrix} \sim \begin{bmatrix} 0 & 0 & 0 \\ 1 & -3 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{cases} x_1 - 3x_2 = 1 \\ x_1 = 1 + 3x_2 \end{cases} \quad v_{a_1} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Then we have the similarity transform:

$$A = P \, \mathcal{J} \, P^{-1}$$

Where:

$$P = \begin{bmatrix} 3 & 1 \\ 1 & 0 \end{bmatrix}, \quad \mathcal{J} = \begin{bmatrix} -3 & 1 \\ 0 & -3 \end{bmatrix}, \quad P^{-1} = \begin{bmatrix} 0 & -1 \\ -1 & 3 \end{bmatrix}$$